

10 Pearls AQI Predictor

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1. Introduction

The 10Pearls AQI Predictor forecasts the Air Quality Index (AQI) for Karachi, Pakistan, over the next 72 hours using a fully serverless ML pipeline. It combines automated data ingestion, feature engineering, model training, and an interactive Streamlit dashboard for visualization.

Key aspects:

- Automated: GitHub Actions for hourly feature updates and daily model retraining.
- Serverless: No dedicated servers required; fully cloud-based with Hopsworks for features and models.
- Scalable: Docker containerization ensures reproducibility and easy deployment.

2. Exploratory Data Analysis (EDA)

Data was collected using the Open-Meteo API, including pollutants and weather data for Karachi (Lat: 24.8607, Lon: 67.0011). EDA was performed on historical air quality and weather data to understand trends and correlations.

2.1 Collected Features:

- Pollutants: PM2.5, PM10, CO, NO₂, O₃, SO₂, NH₃
- Weather: Temperature, humidity, pressure, wind speed, precipitation
- Time: Hour, weekday, month, seasonal encoding
- Derived Features:
 - AQI change rate
 - Lagged AQI values (e.g., previous hour)
 - Rolling averages (3-hour, 24-hour)
 - Wind-pollution interaction
 - Temperature × humidity

2.2 Feature Selection Approach:

- Correlation analysis removed redundant features.
- Rolling AQI averages (especially 3-hour and 24-hour) were most influential.
- Cyclical time encoding improved temporal understanding.
- SHAP values ranked feature importance for interpretability.

- Final features were stored in the Hopsworks Feature Store for model training.

Key Insights:

- AQI spikes occur with low wind speed and high PM2.5.
- Hour-of-day and day-of-week are significant predictors of AQI.
- Weather features (temperature, humidity) moderately correlate with AQI.

3. Feature Pipeline

Implemented in `fetch_features.py`:

- Fetches real-time data hourly from Open-Meteo.
- Processes raw data to generate model-ready features.
- Stores features in Hopsworks Feature Store (versioned).
- Supports historical backfill and real-time ingestion.

4. Model Development & Evaluation

Several models were trained and evaluated to find the best-performing one:

4.1 Models Used

- Classical ML: Random Forest Regressor, Ridge Regression
- Gradient Boosting Regressor (optional experimentation)

4.2 Evaluation Metrics

- RMSE (Root Mean Squared Error)
- MAE (Mean Absolute Error)
- R^2 (Coefficient of Determination)

4.3 Training Process

1. Features and targets fetched from Hopsworks.
2. Models trained and validated using cross-validation.
3. Best model selected and saved to Hopsworks Model Registry.

4.4 Outcome

- Deployed model: Ridge Regression for 72-hour AQI forecasts.
- Performance:

METRIC	VALUE
RMSE	0.0334
R ²	1.0000
R ²	0.0190

- Predictions (Karachi AQI):

DAY	AQI	CATEGORY
1	85.17	Moderate
2	85.68	Moderate
3	86.19	Moderate

- Feature importance: PM2.5, AQI change rate, and hour-of-day were most predictive.

5. Deployment & Visualization

- Streamlit Dashboard (app.py):
 - Visualizes live AQI, pollutant breakdowns, and 72-hour forecasts.
 - Interactive graphs for AQI trends and weather data.
 - Alerts for hazardous AQI levels.
 - SHAP-based plots for feature importance.
- CI/CD Automation:
 - Hourly features update: `.github/workflows/fetch_features.yaml`
 - Daily model retraining: `.github/workflows/train_model.yaml`
- Docker ensures reproducibility and easy deployment.

6. Dashboard Features

- Live AQI + pollutants breakdown
- Graphs: historical and predicted trends
- Color-coded AQI categories: Good → Hazardous
- SHAP interpretability visualization
- Alerts for hazardous conditions

7. Final Evaluation & Limitations

Evaluation:

- Reliable short-term AQI forecasts (RMSE ~ 0.0334).
- Streamlit dashboard provides real-time actionable insights.
- Daily model retraining maintains accuracy.

Limitations:

- Trained specifically for Karachi; other cities require retraining.
- Sudden pollution spikes (fires, traffic) may reduce forecast accuracy.
- Dependent on Open-Meteo API for pollutant data.
- Deep learning models (e.g., LSTM) not deployed; classical models suffice for short-term forecasts.

8. Extras / Additional Features

- Automated alerts for hazardous AQI levels.
- SHAP feature importance for model explainability.
- Fully serverless and reproducible via Docker.

9. Future Improvements

- Online feature lookback during inference via Hopsworks.
- Multi-city AQI forecasting.
- Integration of alert system for hazardous AQI.
- Advanced models: LSTM / Transformer for long-term predictions.
- Expanded explainability with SHAP or LIME.





